

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch: Civil Engineering Semester: VIIth Semester

Subject Code & Name: BTCVE705D Rock Mechanics

Max Marks: 60

Date:11-01-24

Duration: 3.00 Hr.

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

Marks

Q. 1 Solve Any Two of the following.

- A) Write short notes on: - **6**
- a. Lineaments
 - b. Discontinuities in Rocks
 - c. History of rock mechanics
- B) What are its primary objectives and scope in the field of rock mechanics? **6**
- C) Explore the factors that affect the strength of rocks. **6**

Q.2 Solve Any Two of the following.

- A) Write a short note on: - **6**
- a. Primary purpose of rock testing
 - b. Uniaxial compressive strength of rock
 - c. Tri-Axial compressive test in rock testing
- B) Discuss purpose and significance of swelling and slake durability tests. **6**
- C) Explain the importance of tensile Strength of rocks and the methods used for its determination in the laboratory. **6**

Q. 3 Solve Any Two of the following.

- A) Write a short note on: - **6**
- a. Concept of rock mass
 - b. NGI Classification System
 - c. Rock Quality Designation (RQD)
- B) Discuss the factors influencing the behavior of rock masses at slopes. **6**
- C) Explain the Geological Strength Index (GSI) and its significance in rock mass classification. **6**

Q.4 Solve Any Two of the following.

- A)** Write a short note on: - **6**
- a. Rock falls
 - b. Slope monitoring techniques
 - c. Rock slope prevention techniques
- B)** Explain the three main modes of failure in rock slopes with neat labeled diagrams. **6**
- C)** Describe the methods commonly used for the analysis of rock slope stability. **6**

Q. 5 Solve Any Two of the following.

- A)** Write a short note on: - **6**
- a. Rock Reinforcement
 - b. Rock bolting
 - c. Shotcrete
- B)** Discuss the different types of grouting techniques and their applications in enhancing rock mass properties. **6**
- C)** Explain the key differences between the Mohr-Coulomb and Hoek and Brown criteria. **6**

***** End *****